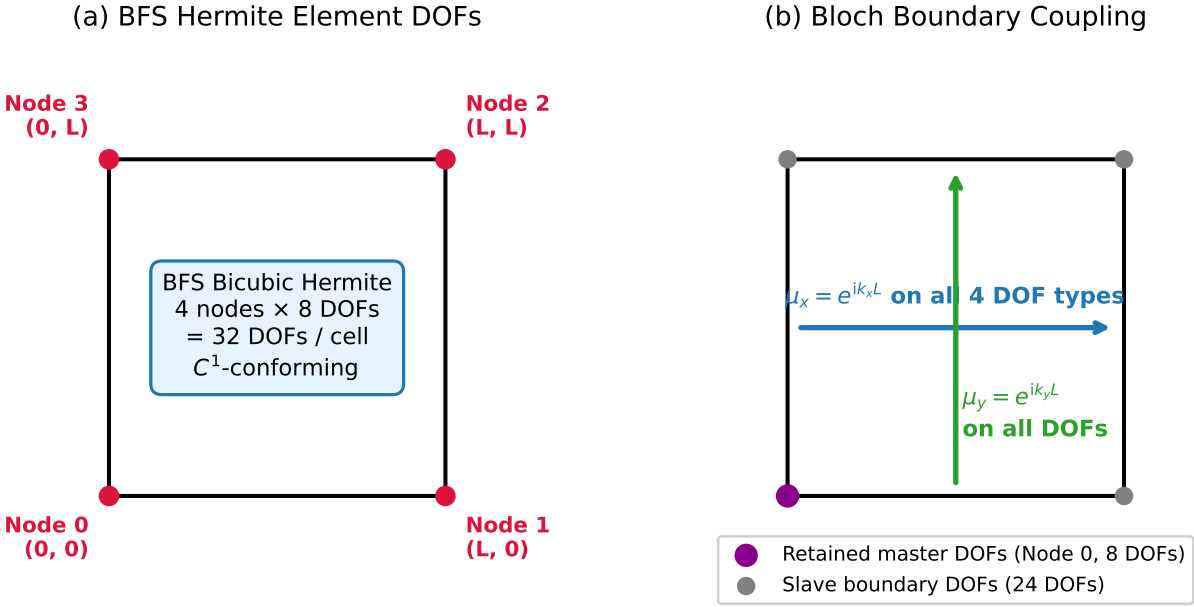
(a) BFS Hermite Element DOFs
Node 3
(0, L)
Node 2
(L, L)
BFS Bicubic Hermite
4 nodes × 8 DOFs
= 32 DOFs / cell
$C^1$-conforming
Node 0
(0, 0)
Node 1
(L, 0)
(b) Bloch Boundary Coupling
$\mu_x = e^{ik_xL}$ on all 4 DOF types
$\mu_y = e^{ik_yL}$
on all DOFs
Retained master DOFs (Node 0, 8 DOFs)
Slave boundary DOFs (24 DOFs)